INTRAC 2000 driver

Overview

The INTRAC 2000 protocol for Adroit communicates to a INTRAC 2000 Central Interface Unit (CIU) via a serial rs232 port that receives data from remote outstations.

The protocol allows both polling for data as well as receiving data on exception (i.e. data is received asynchronously when a value changes at an outstation).

Protocol Manufacturer

Motorola.

Wiring

No	Intrac CIU (25 pin RS-232C male)		Computer Interface (25 pin RS-232C female)
1	2 - Transmit	←→	2 - Receive
2	3 – Receive	←→	3 – Transmit
3	7 – Signal Ground	←→	7 – Signal Ground

Driver Installation and Setup

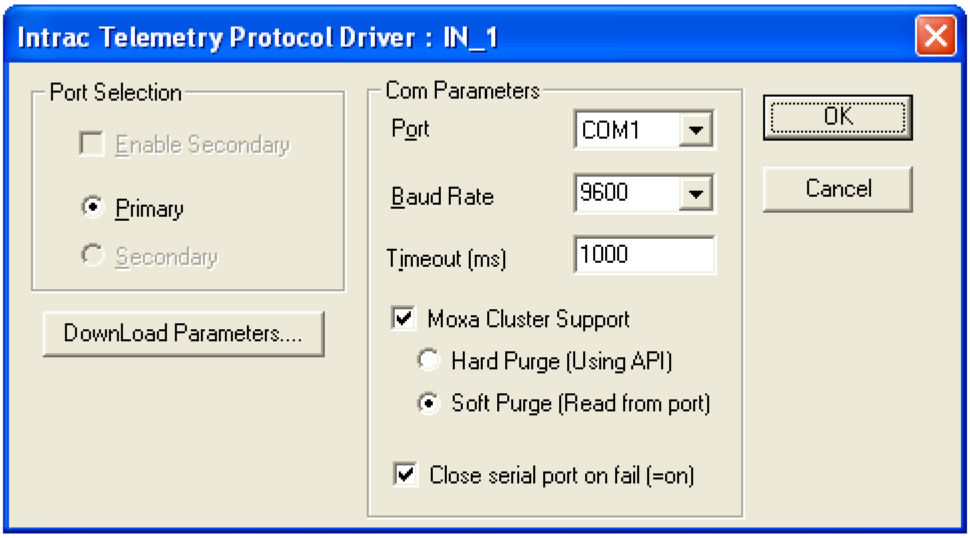
The driver is installed by picking the INTRAC driver from the drivers directory. The INTRAC.DLL and the INTRAC.DOC are then copied into the Adroit directory.

After installation of the driver, the user may then create and configure devices for the INTRAC driver.

Device Configuration

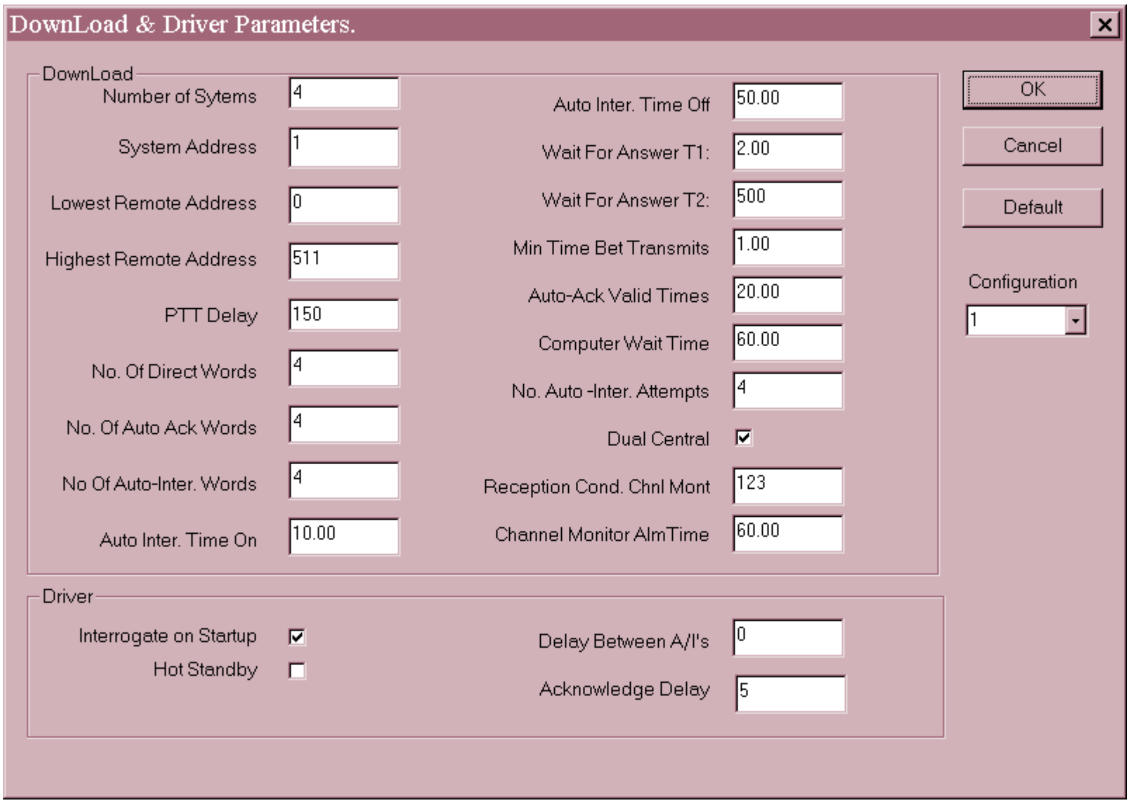
Devices

At a minimum, you must create at least one device for every port being used. However, you may choose to create a device for a group of transmitters. Device configuration dialog



Port	Select the port through which you are going to communicate.
Baud Rate	The baud can be selected. Other parameters are fixed at the standard rate of 7 data, 1 stop, and odd parity.
Timeout	This field gives the waiting time if not end of message is received once a start of message is received. It is recommended that this value be kept at 1000.
Moxa Cluster Support	Check this option when running on a MOXA virtual port in a cluster environment. It will allow to purge the port when in standby and in shadow mode, as the MOXA unit stops functioning in this mode. Data keeps filling the buffer seeing that the driver does not do any reading from the port and eventually becomes unresponsive!
Hard Purge	Does a call to the COM PORT to purge all the data on it. (Preferred Method)
Soft Purge	Continue to read data from the port and just discard it.
Close serial port on fail	With a transaction failure closes the port and re-opens it by default but for some virtual ports it becomes an issue so we allow not closing the port in this case.

Download & Driver Parameters



These Download parameters are explained in the appendix.

Driver Parameters

Interrogate on Startup – All outstations are interrogated on agent server load.

Hot Standby – The driver does not send any data to outstations but only receives data and updates its database. This parameter can be toggled by using system tag number 9.

Delay Between AI's – To slow AI's down – not really required since driver only polls after a RI is received.

Acknowledge Delay – An AA message is sent only after this specified delay , so that more than 1 group can be acknowledged at once.

Scanning

System Tags is: SY: B

Where: SY – System.
B – Bit number (0 - 15)

Boolean Support Only

These bits represent the following:

(Configure Pulsed Items – these bits must be configured as pulsed items as they indicate the condition on a raising edge)

Bit 0 - 'BS' Radio channel busy. - Configure Pulsed
Bit 1 - 'CB' Check back (indication only). - Configure Pulsed
Bit 2 - 'OF' Overflow alarm. Any other message received will clear this bit. It is like a NAK NAK message - Configure Pulsed
Bit 3 - 'RS' Request for restart for CIU. - Configure Pulsed
Bit 4 - 'CM' Channel monitor received. - Configure Pulsed
Bit 5 - 'TX' received after a control.
Bit 6 - 'RI' request interrogation received.
Bit 7 - The CIU is OFF LINE, no CK for 5 minutes.
Bit 8 - Forced CIU download. - Configure Pulsed
Bit 9 - Hot Standby Toggle. - Configure Pulsed
Bit 10 - Force AI even if RI not received (RI outstanding reset)- Configure Pulsed
Bit 11 - Force DW even if TX not received (TX outstanding reset)- Configure Pulsed
Bit 12 – 15 - Unused

Outstation Status Tags: OS: St: Gr: [B:] AI

Where: OS – Outstation.
St – Outstation Number
Gr – Group/s A , BCH , CD , ABCDEFTGH, etc
B – Bit number (0 – 15)
AI – Auto Interrogation rate in minutes, 0 time disables auto interrogation.

Boolean & Integer Support (Multi State).

Bit 0 - AC power failure
Bit 1 – Group A communication failure
Bit 2 - Group B communication failure
Bit 3 - Group C communication failure

Bit 4 - Group D communication failure
Bit 5 - Group E communication failure
Bit 6 - Group F communication failure
Bit 7 - Group G communication failure
Bit 8 - Group H communication failure
Bit 9 – Manual Poll does Auto Interrogate on demand. (Tag must be output enabled)
Bit 10 – 15 – Unused.

Group Data Tags: GD: St: Gr: [B] [T]

Where: GD – Group Data
St – Outstation Number
Gr – Group A,B,C,D,E,F,G or H
B – Bit number – 0- 7
T – Analogue type N or C

Boolean, Integer and Real Support.

For digital tags a maximum of 8 scan points from a Group can be scanned.

For analogues a group can be scanned in as either Normal(N) or as a Counter(C). Scanning of integers made of 2 groups is to be scanned individually and calculated using an expression agent.

Digital Ouput Tags: DO: St: Gr: [B] [T]

Where: DO – Digital Output
St – Outstation Number
Gr – Group A,B,C,D,E,F,G or H
B – Bit number
T – Digital Output type – SM,SL,PN or PR

SM - Single Momentary : the outputs to be toggled are set to 1 and the rest of the bits to 0. These bits are to be configured as pulsed items. The output is only triggered on a rising edge. 8 bits are used for every group.

SL - Single Latched : a latched command ,the outputs to be set on are set to 1 and the outputs to be reset are set to 0. After the transaction the bits remains in their respective states. Outputs are sent both on rising and falling edges.8 bits are used for every group.

PN - Paired Normal : If a bit is set, the data transmitted is a 1 and 0 ,and a bit is reset the data transmitted is a 0 and 1. 4 bits are used for every group.

PR - Paired Reverse : If a bit is set, the data transmitted is a 0 and 1 ,and if a bit is reset the data transmitted is a 1 and 0.

Analogue Ouput Tags: AO: St: Gr: [B] [T]

Where: AO – Analogue Output
St – Outstation Number
Gr – Group A,B,C,D,E,F,G or H

Integer and Real Support.

Message protocol

Datascope messages are displayed in hex or ASCII. All messages are displayed whether that transmitter is configured or not. Messages of devices not configured are displayed in the first devices protocol monitor.

The INTRAC protocol is ASCII. The structure of transactions is as follows:

Read analogue/digital Inputs

Auto Interrogate:

<STX><AI><ADR><AG><RG><ETX><CS><CR><LF>

CIU reply's

<STX><AN><ADR><G><DAT><ETX><CS><CR><LF>

Auto Acknowledge:

(Auto Acknowledge once requested groups data arrives)

<STX><AA><ADR><RG><ETX><CS><CR><LF>

(Acknowledge Messages)

<ACK><ACK><CR><LF>

<NACK><NACK><CR><LF>

Direct Word - Outputs:

<STX><DW><ADR><G><DAT><ETX><CS><CR><LF>

(Download parameters Px where x is 1,2,3,4,5)

<STX><PP><DAT><DAT><DAT><DAT><ETX><CS><CR><LF>

CIU reply's

(Acknowledge Messages)

<ACK><ACK><CR><LF>

<NACK><NACK><CR><LF>

(Spontaneous Data)

<STX><DT><ADR><G><DAT><ETX><CS><CR><LF>

(Failed Groups on Interrogate)

<STX><FL><ADR><AG><ETX><CS><CR><LF>

(Busy/Check/OverFlow/Request Interrogation/Restart)

<STX><FI><ETX><CS><CR><LF>

where:

Field	Bytes	Range	Format	Description
STX	1	02	Binary	Start of Text
ETX	1	04	Binary	End of Text
CR	1	13	Binary	Carriage Return
LF	1	10	Binary	Line Feed
AI	2	'AI'	ASCII	Auto Interrogate
AN	2	'AN'	ASCII	Answer for Inputs
DW	2	'DW'	ASCII	Direct Word for Outputs
PP	2	'P1'..'P5'	ASCII	Download Parameters
FI	2	'BS'	ASCII	Busy

		'CK' 'OF' 'RI' 'RS'		Check Back Overflow Request Interrogation Restart
FL	2	'FL'	ASCII	Failed Groups on interrogate
G	1	'A'..'H'	ASCII	Groups
DAT	8 to 10 bits	0 - 3FF(hex)	ASCII	8 bits per group for 8 digital 10 bits per group for a analogue
AG	1	00 FF	ASCII	Awaited Groups
RG	1	00-FF	ASCII	Requesteg Groups (Same as AG)
CS	2	00-FF	binary	Checksum

Troubleshooting

Appendix

CIU download configuration parameters

Configuration (4, 8, 16).

Default value 4. The CIU can address up to 512 stations out of a total of 2048 different remote stations in an INTRAC 2000 system. The parameter determines the number of systems among which the 2048 remote stations are to be allocated. If 1-4 systems are being used, the value "4" is assigned to this parameter, i.e. maximum 512 stations per system. If 5-8 systems are used, the value "8" is assigned, i.e. maximum 256 stations per system. If 9-16 systems are used, the value "16" is assigned, i.e. maximum 128 stations per system.

System Address (0-3 or 0-7 or 0-15).

Default value 0. Each system assigned to a CIU requires an identification address between 0 and 3. If 2-4 systems are being used, each system must be assigned a separate address between 0 and 3. If 5-8 systems are being used, each system must be assigned a separate address between 0 and 7. If 9-16 systems are being used, each system must be assigned a separate address between 0 and 15. **NB : It is important to assign separate system addresses to adjacent INTRAC 2000 systems using identical frequencies, to avoid inter system interference**.

Lowest Address (0-511 or 0-255 or 0-127)

Default value 0. This parameter determines the lowest remote station address in the system from which the CIU will recognize messages. Messages sent from a remote station with an address lower than the value of this parameter will not be recognized by the CIU.

Highest Address (0-511 or 0-255 or 0-127)

Default value 511 or 255 or 127, depending on system configuration. This parameter determines the highest remote station address in the system from which the CIU recognizes messages sent from a remote station with an address higher than this parameter will not be recognized by the CIU.

PTT Delay (0-635 msec)

Minimal time resolution is 5msec, default value 150msec. This is the amount of time the CIU allows for transmitter warm-up before transferring the first code-word to the transmitter.

Number of Direct Words (0-127)

Default value 4. This parameter determines the number of times the codeword is repeated in each word-burst transmitted to the remote station during a Direct Acknowledgement or Control message transmission.

Number of Auto-Ack Words (0-127)

Default value 4. This parameter determines the number of times the codeword is repeated in each word-burst transmitted to the remote station during an Auto-Acknowledge message transmission.

Number of Auto-Intrg Words (0-127)

Default value 4. This parameter determines the number of times the codeword is repeated in each word-burst transmitted to the remote station during an Auto-Interrogation message transmission.

Auto-Intrg Time On (0-63.5 sec)

Minimim time resolution is 0.5sec, default value 10 sec. This parameter determines the amount of time per on/off cycle that the CIU is enabled for transmission during Auto-Interrogation. The value of both this parameter and the Auto-Intrg Time-Off parameter are determined after considering the relevant government regulations and the Auto-Interrogation cycle dynamics for the system database, as outlined in Section 9 of the CIU Operating Instructions Manual.

Auto-Intrg Time Off (0-63.5 sec)

Minimim time resolution is 0.5sec, default value 50 sec. This parameter determines the amount of time per on/off cycle that the CIU does not transmit during Auto-Interrogation. The value of both this parameter and the **Auto-Intrg Time-On** parameter are determined after considering the relevant government regulations and the Auto-Interrogation cycle dynamics for the system database, as outlined in Section 9 of the CIU Operating Instructions Manual.

Wait-for-Answer T1 (0-6.35 sec)

Minimal time resolution is 0.05sec, default value 2 sec. After each autonomous transmission of interrogation code-words, the CIU waits a predetermined period of time for an answer from the interrogated station. During this period the CIU makes no additional automatic interrogations. The Wait-for-Answers T1 and Wait-for-Answers T2 parameters determine the actual length of time that the CIU waits before indicating that the remote station being interrogated has failed. These two parameters are explained more fully in Section 9.5 in the CIU Operating Instructions Manual.

Wait-for-Answer T2 (0-635 msec)

Minimim time resolution is 5 msec, default value 500 msec. After each autonomous transmission of interrogation code-words, the CIU waits a predetermined period of time for an answer from the interrogated station. During this period the CIU makes no additional automatic interrogations. The **Wait-for-Answer T1** and **Wait-for-Answer T2** parameters determine the actual length of time that the CIU waits before indicating that the remote station being interrogated has failed. (See Section 9.5 of the CIU Operating Instructions Manual).

Min. Time between TX (0-6.35 sec)

Minimim time resolution is 0.05sec, default value 1 sec. This parameter determines the minimum interval between any two consecutive CIU transmissions, if required (either by government regulations or by equipment requirements).

Auto-Ack Valid Time (0-63.5 sec)

Minimim time resolution is 0.5sec, default value 20 sec. This parameter determines the maximum amount of time the CIU waits for a free communication channel before cancelling an automatic acknowledgment request.

Computer Wait Time (0-63.5 sec)

Minimim time resolution is 0.5sec, default value 60.0 sec. This parameter determines the maximum time the CIU devotes to transferring a message to the computer or to receiving from the computer an acknowledgment that the message was received.

Auto-Intrg Attempts (0-127)

Default value 4. This parameter determines the number of interrogation word-bursts the CIU transmits to a remote station during automatic interrogation while it waits for all of the groups at that station to respond.

Dual Central (Y/N)

Default value Y. This parameter allows the use of more than one central station in the system. See Section 14.9 in the CIU Operating Instructions Manual for an explanation of primary mode and secondary mode central units.

Reception Conditioned with Channel Monitor (0 -127)

If this value is 123 (7B hex) the CIU reception is conditioned with the channel monitor, all incoming messages are filtered for reception of only those messages received while a carrier is present. When the CIU reception is not conditioned (any other value), all messages received by the CIU are forwarded to the central unit. It is sometimes useful to disable the channel monitor during times of poor transmission conditions to prevent the filtering out of valid messages merely because of poor channel monitor operation.

Channel Monitor Alarm Time (0-63.5 sec).

Minimim time resolution is 0.5 sec, default value 60 sec. This parameter determines the length of channel monitor busy time allowed before a "CHANNEL MONITOR" ALARM message is sent to the computer. If the reception conditioned value is set to between 01 - 1D then a factor for the channel monitor alarm parameter is set. The channel monitor alarm is now equal to a value set here multiplied by the above factor. The resultant value now ranges from or 0-1841.5 sec